

California Housing Price Prediction

Linear Regression: Exploratory Analysis, Model & Evaluation

1. Overview

This report summarizes a linear regression workflow built on the California Housing dataset (20,640 records, 8 numeric features, sourced from scikit-learn). The goal is to predict the median house value (in units of \$100,000) for a census block group from demographic and geographic attributes such as median income, house age, average rooms, population, and location.

2. Dataset & Exploratory Data Analysis

The dataset contains 8 predictor features (MedInc, HouseAge, AveRooms, AveBedrms, Population, AveOccup, Latitude, Longitude) and one target (MedHouseVal). A check confirmed there are no missing values in any column, so no imputation was required.

2.1 Correlation Analysis

A correlation heatmap was generated to understand relationships between features and the target. Median income (MedInc) shows the strongest positive correlation with house value, while features like AveBedrms, Population, and AveOccup show weak correlation with the target, hinting that income is the dominant predictive signal.

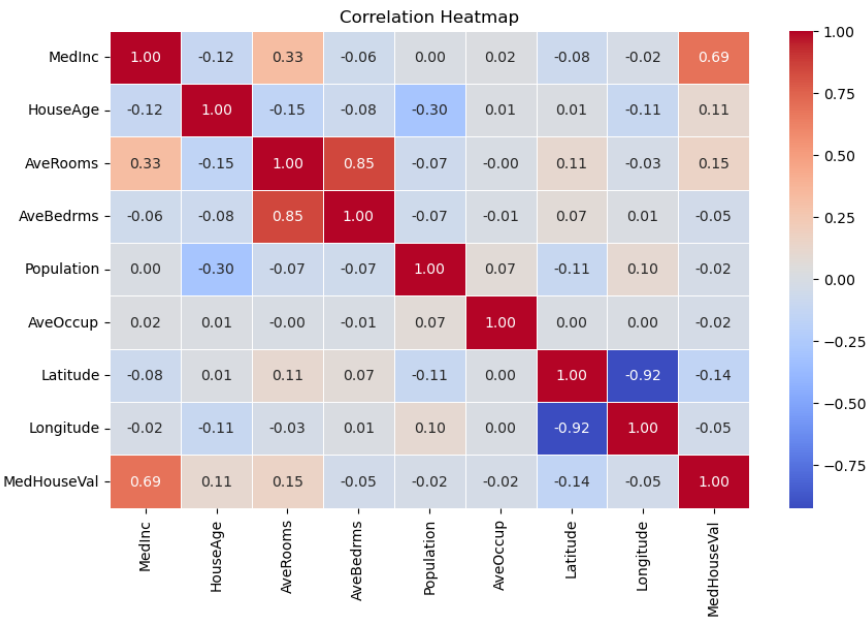


Figure 1: Correlation heatmap of all features and the target variable.

2.2 Feature Distributions

Histograms of all features and the target were plotted to inspect their distributions. Several features (Population, AveOccup, AveRooms, AveBedrms) are heavily right-skewed with long tails, and both MedHouseVal and HouseAge show a noticeable spike at their upper bound, indicating value-capping in the source data (house

values are capped near \$500,000).

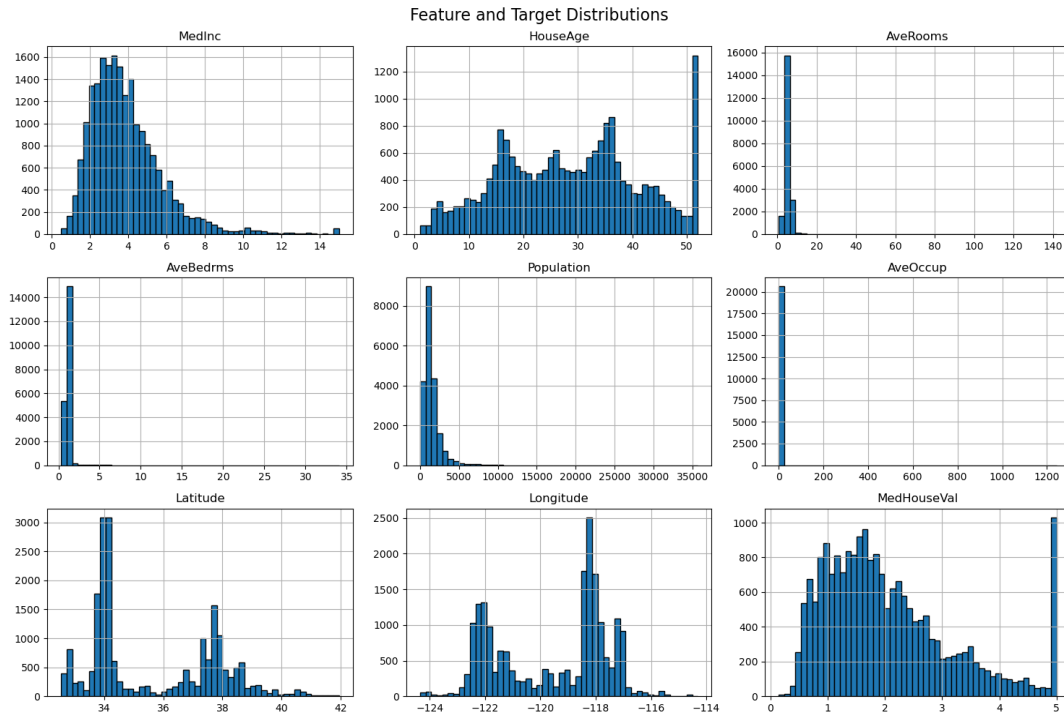


Figure 2: Distributions of all features and the target variable.

3. Modeling Approach

- Split the data into 80% training (16,512 rows) and 20% testing (4,128 rows) using a fixed random seed (42) for reproducibility.
- Standardized all 8 features with **StandardScaler**, fitting only on the training set and transforming both train and test sets to prevent data leakage.
- Trained a simple **Ordinary Least Squares Linear Regression** model (scikit-learn's LinearRegression) on the scaled training data.
- Serialized the trained model and scaler to disk (california_model.pkl, california_scaler.pkl) for reuse.
- Built a lightweight **Streamlit** web app (app.py) that loads the saved model/scaler and lets a user enter neighborhood attributes to get an estimated house price interactively.

4. Model Evaluation

The model was evaluated on the held-out 20% test set using three standard regression metrics:

Metric	Value	Interpretation
MAE (Mean Absolute Error)	0.5332	On average, predictions are off by ~\$53,320
RMSE (Root Mean Squared Error)	0.7456	Typical error magnitude ~\$74,560, penalizing large misses more
R ² (Coefficient of Determination)	0.5758	The model explains ~57.6% of the variance in house values

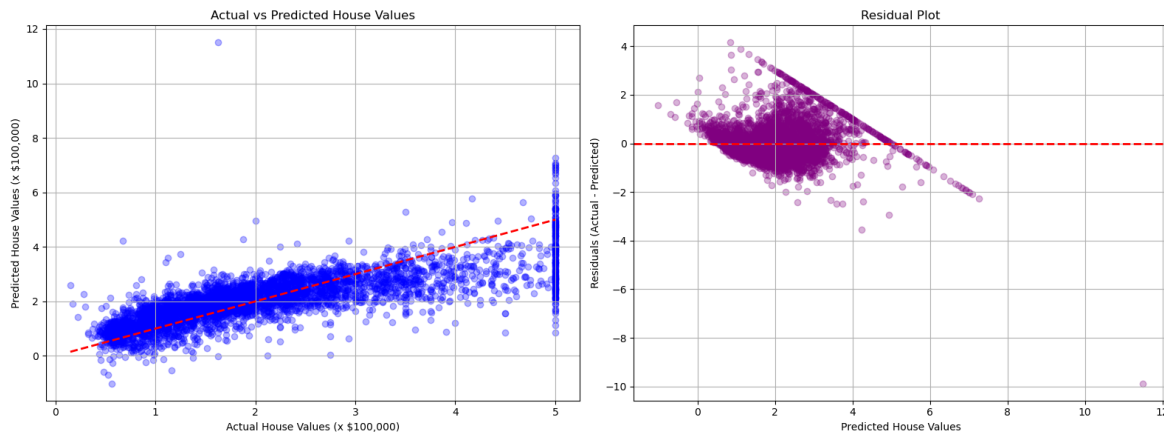


Figure 3: Actual vs. predicted house values (left) and residuals plot (right).

The actual-vs-predicted plot shows a reasonable linear trend, but with visible scatter, especially at higher home values, where the model tends to under-predict. The residual plot shows a cone / fan shape (heteroscedasticity) and a flat ceiling of points near the top-right, consistent with the capped MedHouseVal values seen in the histograms — the model cannot predict beyond what a linear fit allows, while real values are capped at \$500,000.

5. Limitations & Improvement Ideas

Non-linear models: Try tree-based models (Random Forest, Gradient Boosting/XGBoost) or polynomial features, since the relationship between income/location and price is unlikely to be purely linear.

Feature engineering: Derive ratios such as rooms-per-household or bedrooms-per-room, and bin/transform Latitude & Longitude (e.g. distance to coast/major cities, or KMeans-based location clusters) to better capture geographic effects.

Handle the capped target: Since MedHouseVal is capped at \$500,000, consider removing or separately flagging capped rows, or evaluating with metrics that are robust to this ceiling effect.

Address skewed features: Apply log-transforms to heavily skewed features (Population, AveOccup, AveRooms, AveBedrms) to reduce the influence of outliers and stabilize variance.

Outlier handling: Cap or remove extreme values in AveRooms/AveOccup/Population, which can otherwise dominate the least-squares fit.

Regularization: Use Ridge/Lasso/ElasticNet regression to reduce variance and improve generalization, especially if more engineered features are added.

Cross-validation: Replace the single train/test split with k-fold cross-validation for a more robust estimate of generalization performance.

Residual diagnostics: Investigate the heteroscedasticity seen in the residual plot further — e.g. via weighted least squares or a variance-stabilizing transform on the target.

6. Conclusion

The baseline linear regression model provides a reasonable but limited fit ($R^2 \approx 0.58$) on the California Housing dataset, driven largely by median income. The residual and distribution analysis point to non-linearity, a capped target variable, and skewed features as the main factors limiting performance — all addressable with the improvement ideas above before moving to a production-grade model.